1 The query S: two seeds

Samsung SATA optical drive
... Optical Drives > DVD Drives $f = 1$

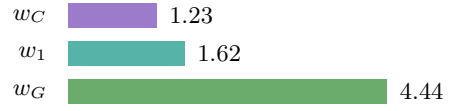
Lite-On 24X SATA DVD drive
... > Internal DVD Drives $f = 4$

Missing item (the target): Seagate Barracuda 320 GB internal hard drive, category ... Data Storage > Internal Hard Drives, $f = 11$.

2 The gate: three numbers about S become three weights

$u(S) = \big(\log|S|,\ \overline{\log(1+f_s)},\ \text{share unseen}\big)$
$= (0.69,\ 1.15,\ 0)$

MLP, softplus



Both seeds were seen in training, so the gate trusts co-occurrence most: $w_G \approx 3.6\, w_C$.

3 The three terms for the target c = Seagate hard drive

content $C(S,c)$
$\varphi(S) = \frac{1}{2}\big(\text{query}(s_1) + \text{query}(s_2)\big)$
lengths 2.76, 2.78; $\|\varphi(S)\| = 2.68$,
the seeds point the same way
$C = \varphi(S)^\top \text{key}(c) = \mathbf{5.18}$

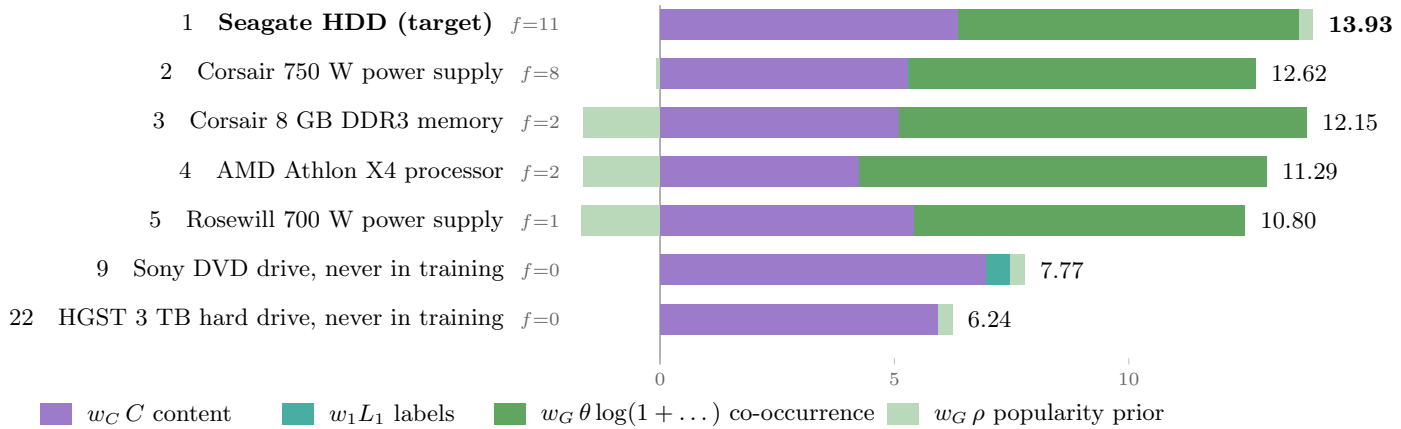
labels $L_1(S,c)$
no seed shares the target's category:
$m = 0$, so the diagonal δ_1 is off
low rank:
$\frac{1}{|S|} t_1(S)^\top P_1 M_1 Q_1 [g_1(c)]^\top$
$= \mathbf{-0.005}$, near zero

graph $G(S,c)$
$N_{s_1 c} = 1$, $N_{s_2 c} = 2$; $\deg_{s_1} = 7$,
$\deg_{s_2} = 13$, $\deg_c = 52$
$\alpha = 0.036$, $\beta = 0.175$:
$\sum_s N_{sc} \deg_s^{-\alpha} \deg_c^{-\beta} = 1.38$
$\theta \log(1 + 1.38) = 1.89 \times 0.87 = 1.64$
$\rho(\log 12) = 0.06$, so $G = \mathbf{1.71}$

4 The score: weighted sum, for every item at once

$s(c \mid S) = w_C\, C + w_1 L_1 + w_G\, G = 1.23 \times 5.18 \ +\ 1.62 \times (-0.005) \ +\ 4.44 \times 1.71 = \mathbf{13.93}$ → rank 1 of 3,497

5 Where each candidate's score comes from



Bars: each part's weighted contribution to $s(c \mid S)$; a negative prior is drawn left of zero. Label contributions under 0.02 in size are not drawn. Frozen Electronic model, seed 0, test query 20; the scores rank all 3,497 non-seed items.